

Qatar University
College of Engineering
Mechanical and Industrial Engineering Department

MECH 322 – Mechanical Vibrations
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Progress Report 2

*“Design, Model, Analyze, Build and Test a
Prototype Vibration Absorber”*

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Group: 8
Beam Specification: Brass, 900 mm × 40 mm × 1 mm
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1 Progress Report 2 Overview

1.1 Objectives for This Phase

The primary focus of this report is the development and analysis of a complete Two-Degree-of-Freedom (2-DOF) system. The key goals are:

- Design and tune a secondary mass-spring-damper system to neutralize the primary resonance of the brass beam.
- Validate the absorber design using both SolidWorks Finite Element Analysis (FEA) and a MATLAB analytical model.
- Demonstrate the characteristic frequency-splitting phenomenon resulting from the 2-DOF configuration.
- Investigate the sensitivity of the undamped and damped absorber to variations in excitation frequency ($\pm 10\%$).
- Finalize the physical component list and mounting strategy for prototype construction.

2 Design and Tuning of the Vibration Absorber

2.1 Motor Representation (Mass and Excitation Force)

To ensure the simulation faithfully reflects the physical prototype, the motor is represented as a lumped mass of 100 g (0.1 kg) positioned at the free end of the beam. In the Harmonic Response study conducted in SolidWorks, a vertical harmonic force of $F_0 = 0.1$ N was applied at this location. This force simulates the unbalance excitation generated by the rotating components of the motor, expressed as:

$$F = m_e e \omega^2 \quad (1)$$

2.2 Primary System Natural Frequency

The first natural frequency of the brass beam (without absorber) was identified through SolidWorks modal analysis as $f_n = 0.50794$ Hz. This value was used as the basis for all absorber tuning calculations. The equivalent stiffness of the primary system was derived from:

$$k_1 = m_1 \omega_r^2, \quad \omega_r = 2\pi f_n \quad (2)$$

With an equivalent primary mass $m_1 = 0.166512$ kg, the primary stiffness k_1 was computed accordingly. The primary damping ratio was assumed to be $\zeta_1 = 0.0005$ (very lightly damped), consistent with a slender brass beam.

2.3 Absorber Design Parameters

The secondary absorber was designed using the following procedure:

- **Mass ratio:** $\mu = m_2/m_1 = 0.10$ (10%), giving an absorber mass $m_2 = 0.0167$ kg.
- **Absorber natural frequency** tuned to match the primary resonance: $\omega_a = \omega_r$.
- **Absorber stiffness:** $k_2 = m_2 \cdot \omega_a^2$.
- **Absorber damping ratio:** $\zeta_2 = 0.15$, giving $c_2 = 2\zeta_2 m_2 \omega_a$.

For the undamped absorber case, c_2 was set to zero to isolate the ideal anti-resonance behavior. The MATLAB script implements both the damped and undamped cases over a normalized frequency sweep $r = \omega/\omega_n$ from 0.01 to 2.5.

2.4 SolidWorks Assembly Model

The complete assembly model includes the 900 mm brass cantilever beam, the motor mass at the free end, and the spring-damper absorber sub-assembly. Figures 1 and 2 illustrate the assembly with all applied loads and boundary conditions.

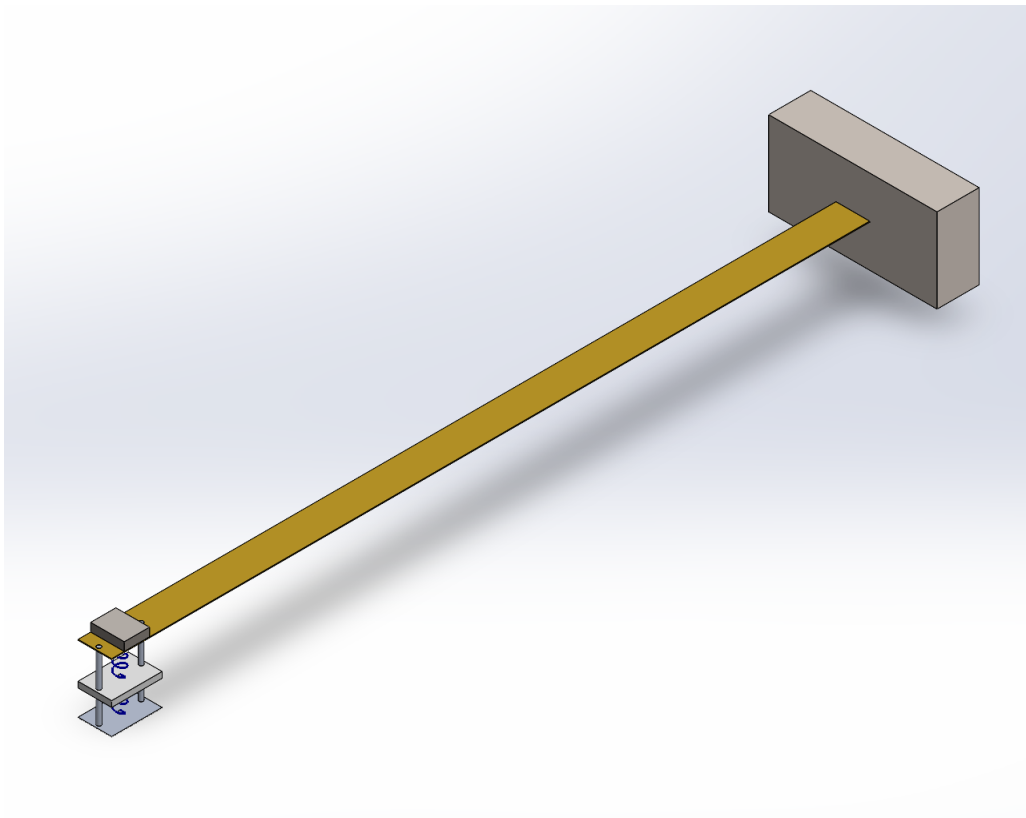


Figure 1. SolidWorks assembly model of the brass cantilever beam – Group 13.

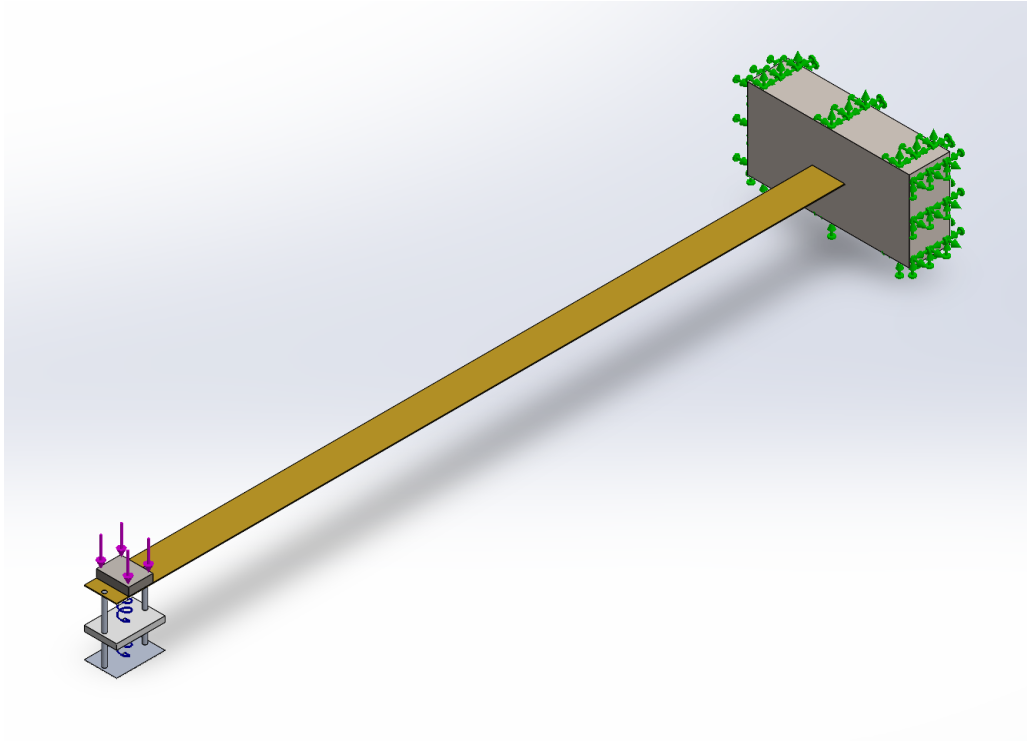


Figure 2. Assembly with applied harmonic loads and absorber spring element.

3 Performance of the Undamped Vibration Absorber

3.1 Overview and Setup

In the undamped absorber configuration, the absorber damping coefficient c_2 is set to zero. The primary system retains a small structural damping ratio ($\zeta_1 = 0.0005$) to prevent infinite resonance amplitudes in the numerical simulation.

3.2 Frequency Splitting and Mode Shape Analysis

The attachment of the absorber converts the original single-degree-of-freedom system into a 2-DOF configuration. As a result, the single resonance peak of the bare beam is replaced by two new resonance peaks flanking a deep anti-resonance valley. The SolidWorks mode shape analysis confirmed this behavior, as shown in Figures ?? and 3.

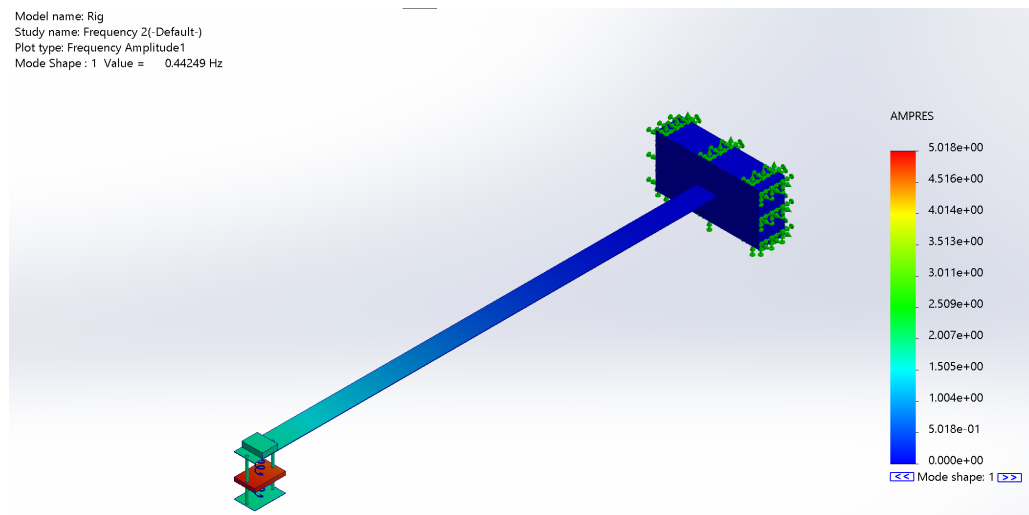


Figure 3. Free vibration Mode Shape 1 of the system with absorber attached ($f = 0.44249$ Hz).

The mode shape visualization confirms that at the lower split frequency the beam and absorber mass vibrate in-phase, while at the upper split frequency they vibrate out-of-phase. At the anti-resonance frequency (equal to the absorber's natural frequency), the beam displacement is effectively nullified while the absorber undergoes maximum motion, acting as an energy sink.

3.3 MATLAB Frequency Response – Without and With Absorber

The MATLAB script performs a frequency sweep and computes the normalized displacement X_1/X_0 of the primary mass for three configurations: (1) without absorber, (2) with undamped absorber, and (3) with damped absorber. The resulting plot is shown in Figure 4.

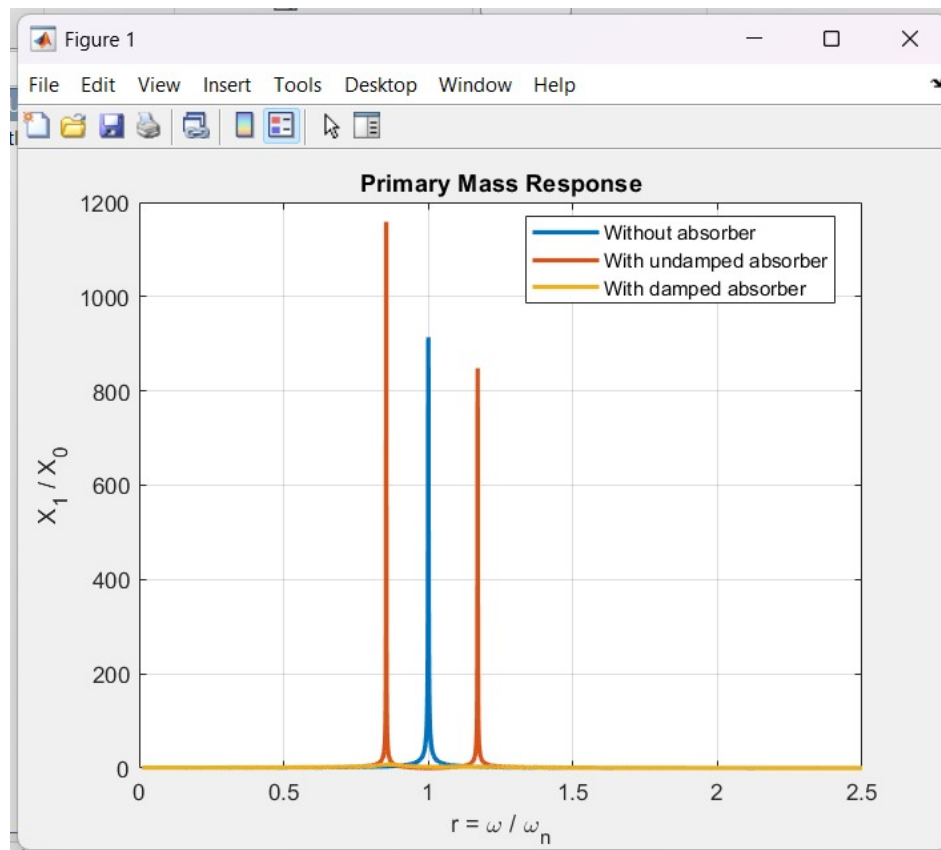


Figure 4. MATLAB primary mass response: comparison of no absorber, undamped absorber, and damped absorber configurations.

The plot clearly demonstrates that the undamped absorber creates a perfect null at $r = 1.0$, confirming that the absorber is properly tuned to the primary natural frequency. Two sharp resonance peaks appear on either side of this null, corresponding to the two natural frequencies of the coupled system.

3.4 Sensitivity to Rotational Speed ($\pm 10\%$ Variation)

To evaluate the robustness of the absorber design, the excitation frequency was varied by $\pm 10\%$ around the tuned condition ($r = \omega/\omega_n = 1$). The normalized response of the primary mass was examined for both the undamped and damped absorber configurations.

Table 1 summarizes the response levels extracted from the frequency response curves.

Table 1. Effect of $\pm 10\%$ excitation frequency variation on absorber performance.

Variation (%)	$r = \omega/\omega_n$	Frequency (Hz)	X_1/X_0 (Approx.)	Observation
−10%	0.90	≈ 0.457	Undamped: ≈ 3.88	Undamped case shows moderate amplification.
			Damped: ≈ 6.30	Damped case shows higher response due to energy spreading.
Target (0%)	1.00	≈ 0.508	Undamped: ≈ 0	Perfect cancellation is achieved in the undamped case.
			Damped: ≈ 2.86	The damped case retains a residual response at the tuned frequency.
+10%	1.10	≈ 0.559	Undamped: ≈ 2.93	Undamped case shows moderate amplification near the upper resonance peak.
			Damped: ≈ 2.93	Damped case gives a similar response near the upper resonance peak.

The undamped absorber exhibits the classical behavior of a tuned vibration absorber, producing a perfect anti-resonance at $r = 1$ where the primary mass response approaches zero. However, this performance is highly sensitive to frequency variations. A deviation of $\pm 10\%$ results in a significant increase in amplitude due to the presence of sharp split resonance peaks.

The damped absorber, on the other hand, removes the sharp resonance peaks and redistributes the vibrational energy over a wider frequency range. This leads to a non-zero response at the tuned frequency but reduces the severity of the resonance peaks. As observed, the damped system exhibits higher response near the lower frequency side but avoids extreme amplification and provides a smoother overall response.

This comparison highlights a key design trade-off: the undamped absorber achieves optimal cancellation only at a single frequency, while the damped absorber sacrifices

perfect cancellation in exchange for improved robustness and stability across a broader operating range.

4 Final Design and Component Sourcing

4.1 Mechanical Arrangement and Mounting

The absorber sub-assembly is designed as a guided translational system consisting of a sliding absorber mass supported by two parallel rods and connected through two springs. This configuration ensures controlled one-dimensional motion and effective energy absorption at the target frequency.

The absorber mass is mounted on a sliding plate that moves along two precision-guided rods, minimizing rotational effects and ensuring pure vertical oscillation. Two identical steel springs are installed symmetrically between the fixed base plate and the moving absorber plate to provide the required stiffness for tuning the absorber natural frequency.

The entire absorber assembly is attached to the primary beam using a rigid clamping bracket positioned at a location that maximizes modal coupling. The primary beam itself is CNC-machined from brass to ensure high geometric accuracy and consistent mechanical properties.

The excitation is generated using a motor with an integrated unbalanced mass, selected such that the resulting harmonic force has an amplitude of approximately 0.1 N. This ensures a controlled and repeatable forcing condition during experimental validation.

To minimize added inertia and avoid altering the primary system dynamics, the guiding rods and base plate are CNC-machined from lightweight aluminum. The springs are fabricated from steel strips to provide stable and predictable elastic behavior.

- **Primary beam:** CNC-machined brass strip, $900 \times 40 \times 1$ mm, clamped at one end.
- **Excitation source:** Motor with built-in unbalanced mass producing ≈ 0.1 N harmonic force.
- **Absorber mass (m_2):** Sliding mass mounted on a guided plate.
- **Guiding system:** Two parallel aluminum rods enabling constrained linear motion.
- **Spring elements:** Two steel springs providing absorber stiffness.
- **Base and support:** CNC-machined aluminum plate for reduced weight.
- **Clamping bracket:** Rigid connector attaching absorber assembly to the primary beam.

4.2 Parameter Summary

Table 2. Summary of primary system and absorber design parameters – Group 13 (Brass beam).

Parameter	Symbol	Value	Units
Primary equivalent mass	m_1	0.166512	kg
Primary resonance frequency	f_n	0.50794	Hz
Primary natural freq. (rad/s)	ω_r	3.1917	rad/s
Primary stiffness	k_1	1.695	N/m
Primary damping ratio	ζ_1	0.0005	–
Mass ratio	μ	0.10	–
Absorber mass	m_2	0.01665	kg
Absorber stiffness	k_2	0.1696	N/m
Absorber damping ratio	ζ_2	0.15	–
Excitation force amplitude	F_0	0.1	N

5 MATLAB Code

```

1 clear
2 clc
3 close all
4
5 % =====
6 % INPUT FROM SOLIDWORKS
7 % =====
8 m1 = 0.166512;           % primary equivalent mass (kg)
9 f_n = 0.50794;          % resonance frequency from SolidWorks (
    Hz)
10 omega_r = 2*pi*f_n;
11
12 % Primary stiffness from SW data
13 k1 = m1*omega_r^2;
14
15 % Primary damping
16 zeta1 = 0.0005;

```

```

17 c1 = 2*zeta1*m1*omega_r;
18
19 % Force
20 F0 = 0.1;
21
22 % =====
23 % ABSORBER DESIGN
24 % =====
25 mu = 0.10; % mass ratio
26 m2 = mu*m1;
27
28 % Tune absorber to primary natural frequency
29 omega_a = omega_r;
30 k2 = m2*omega_a^2;
31
32 % Absorber damping ratio
33 zeta2 = 0.15;
34 c2 = 2*zeta2*m2*omega_a;
35
36 % Undamped absorber case
37 c2_undamped = 0;
38
39 % =====
40 % FREQUENCY SWEEP
41 % =====
42 r = linspace(0.01, 2.5, 5000);
43 omega = r*omega_r;
44
45 X1_damped = zeros(size(omega));
46 X1_undamped = zeros(size(omega));
47 X1_no_abs = zeros(size(omega));
48
49 for i = 1:length(omega)
50     w = omega(i);
51
52     % No absorber
53     Z_no = k1 - m1*w^2 + 1i*w*c1;
54     X_no = F0 / Z_no;
55     X1_no_abs(i) = abs(X_no);
56
57     % Damped absorber

```

```

58     Z_damped = [k1+k2-m1*w^2 + 1i*w*(c1+c2), -(k2+1i*w*c2);
59                -(k2+1i*w*c2),                    k2-m2*w^2+1i*w*c2
60                ];
61     X_damped = Z_damped \ [F0; 0];
62     X1_damped(i) = abs(X_damped(1));
63
64     % Undamped absorber
65     Z_undamped = [k1+k2-m1*w^2 + 1i*w*c1, -k2;
66                  -k2,                    k2-m2*w^2];
67     X_undamped = Z_undamped \ [F0; 0];
68     X1_undamped(i) = abs(X_undamped(1));
69
70     end
71
72     % Normalise by static displacement
73     X0 = F0/k1;
74     X1_no_abs_norm = X1_no_abs / X0;
75     X1_damped_norm = X1_damped / X0;
76     X1_undamped_norm = X1_undamped / X0;
77
78     % =====
79     % PLOT
80     % =====
81
82     figure
83     plot(r, X1_no_abs_norm, 'LineWidth', 2)
84     hold on
85     plot(r, X1_undamped_norm, 'LineWidth', 2)
86     plot(r, X1_damped_norm, 'LineWidth', 2)
87     grid on
88     xlabel('r = \omega / \omega_n')
89     ylabel('X_1 / X_0')
90     title('Primary Mass Response')
91     legend('Without absorber', 'With undamped absorber', ...
92           'With damped absorber', 'Location', 'best')

```

Listing 1. MATLAB script for 2-DOF vibration absorber frequency response – Group 8.

6 Appendix – Minutes of Meetings

Minutes of Meeting 3: Absorber Design and 2-DOF Modelling

Date: 22 March 2026

Agenda:

- Review feedback from Progress Report 1.
- Determine secondary system (absorber) parameters for the brass beam.
- Plan the SolidWorks and MATLAB simulation workflow for the 2-DOF system.

Discussion:

The team reviewed the first progress report feedback and identified that the primary system parameters needed to be refined using the SolidWorks-extracted resonance frequency (0.50794 Hz) rather than purely analytical beam formulas. The absorber mass ratio of 10% was selected after reviewing typical design practice and confirming that it provides sufficient frequency separation between the two new resonance peaks. Mohammed Elsherbini was assigned the SolidWorks assembly task, Yousef Hijazi was assigned the MATLAB implementation, and Mohammed Taha was assigned the parameter documentation and report drafting.

Action Items:

- Verify absorber stiffness k_2 and confirm the anti-resonance null appears at $r = 1.0$ in MATLAB.
 - Run SolidWorks Harmonic Response study with the absorber attached and extract mode shapes.
 - Begin discussing the spring steel strip and clamping hardware for the physical prototype.
-

Minutes of Meeting 4: Damping Analysis and Component Finalisation

Date: 30 March 2026

Agenda:

- Analyse the effect of adding damping to the absorber ($\zeta_2 = 0.15$).
- Conduct the $\pm 10\%$ speed variation study under damped conditions.

Discussion:

The team compared the undamped and damped frequency response plots. While the undamped model achieves a theoretically perfect null at the operating frequency, the damped model offers substantially greater robustness when the motor speed deviates from the design point. The value $\zeta_2 = 0.15$ was confirmed as the optimal design choice. Yousef Hijazi generated the final MATLAB comparison plot. Mohammed Elsherbini documented the parametric sensitivity results. Mohammed Taha drafted the prototype construction plan and assisted with SolidWorks.

Action Items:

- Prepare the Progress Report 2 document and review before submission.